World Organisation for Animal Health | Organisation mondiale de la santé animale | Organización Mundial de Sanidad Animal

https://doi.org/10.20506/rst.41.1.3307

Movement of genetically modified insects for research purposes

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Summary

Insects play a crucial role in research. Many laboratories are developing technologies to control insect vectors or agricultural pests by using genetic modifications that either reduce insect reproduction or increase refractoriness to disease transmission. Those tools include gene-drive elements that may spread such genetic traits in a self-sustaining and cost-effective manner. Since international research collaborations are nowadays routine, movement of genetically modified insects between laboratories under different regulatory jurisdictions is very common. This article describes the requirements and guidelines for transportation of genetically modified insects for research. The author draws upon the experience gained by an Italian laboratory, in its role as a research centre involved in shipments of wild and modified mosquitoes, within an international research consortium developing sustainable tools for malaria control.

Keywords

Arthropod containment – Gene drive – Genetically modified insects – Insects – Transportation.

Introduction

It is difficult to overestimate the importance of the role that insects play in research. Since the days of Thomas Hunt Morgan, who worked with fruit flies (*Drosophila melanogaster*) in the 'fly room' at Columbia University at the beginning of the 20th century, insects have filled many laboratories around the world. The sharing and movement of strains of laboratory insects, such as the thousands of *Drosophila* strains and mutants (to cite the most famous insect model organism), is a very common practice among researchers, as reported by Cook and Parks [1]. With the advent of transgenic technologies, the number of different mutants and genetically modified (GM) strains has dramatically increased, and thus so has the need for movement of strains between laboratories, in a spirit of shared knowledge, transparency and collaboration.

In the last two decades, the number of insect species that have been genetically modified has extended beyond the world of *Drosophila* to several classes of arthropods, which are now routinely modified. Several agricultural pests are reared and transformed in laboratories, including the Mediterranean fruit fly or medfly (*Ceratitis capitata*), the diamondback moth (*Plutella xylostella*), the pink bollworm (*Pectinophora gossypiella*) and the spotted wing Drosophila (*D. suzukii*). Public health research aimed at developing tools to control insect vectors of human or animal diseases is concentrated on *Anopheles* and *Aedes* mosquitoes, the former being responsible for transmitting malaria and filariasis while the latter is a vector for dengue, Zika, yellow fever and chikungunya viruses.

Gene editing and genetic manipulation of insects are efficient tools that have increased our understanding of the molecular and cellular basis of many biological, evolutionary and developmental processes. Over the last decade, innovative genetic technologies have also opened up the possibility of providing solutions to certain medical or agricultural problems caused by pest insects. The novel properties of these insects mean that precautionary containment in laboratories is required while biosafety studies are being conducted. In addition, rearing and maintaining such potential disease vectors must be preceded by an evaluation of the risks of their escape from containment, to alert stakeholders of any other precautions that must be taken in their handling, rearing and especially shipping between laboratories.

Genetically modified insects are defined as organisms in which genetic material has been purposely altered through genetic engineering in a way that does not occur naturally. This includes both stable germline transformation, which targets the nuclei of developing germ cells for DNA introduction, and gene-editing applications that generate mutations through the use of specific nucleases, such as the CRISPR/Cas system (i.e. clustered, regularly interspaced, short palindromic repeats/CRISPR-associated system) [2].

There are several methods for introducing foreign DNA into the developing embryo (or adult ovaries), with the most effective being micro-injection into individual embryos. Once injected into the embryo, DNA must recombine in the target genome by the use of transposable elements, site-specific integration systems, virus-mediated recombination or by homologous recombination. Identifying and selecting stable germline transformation usually relies on phenotypical markers, such as eye or body colour mutants or the use of fluorescent proteins.

One class of genetic modification needing a special mention is gene drive. Gene drive refers to a phenomenon in which heritable elements (genes) bias their own inheritance with respect to the rest of the genome, thereby increasing in frequency in the population ('driving it') over generations [3]. Gene drives are still at the research stage in laboratories and none has yet been released into the environment. Gene drives offer great potential as a tool to control insect disease vectors because their invasive nature allows them to spread and modify entire populations in a limited time frame, making them attractive for cost-effective and self-sustaining public health applications [4]. Since gene drives are designed to be highly effective at establishing in wild populations, guidelines and recommendations are in place to ensure that laboratories conducting research on gene drives meet containment standards to minimise the risk of accidentally releasing gene-drive-carrying organisms [5, 6, 7].

Shipment of genetically modified mosquitoes

Import and export of live mosquitoes for research

Polo d'Innovazione Genomica Genetica e Biologia (Polo GGB) is an Italian research and development laboratory and a service facility for genomics, bioinformatics, genetics and ecology studies. Polo GGB is also a member of Target Malaria, an international research consortium developing genetic technologies to control *Anopheles* mosquitoes, with the aim of reducing the burden of malaria in Africa (https://targetmalaria.org). This technology is based on the use of GM mosquitoes to reduce the population of human malaria mosquitoes and, as a result, reduce malaria transmission. As part of this development pathway, Polo GGB has been involved in several shipments of live mosquitoes between partner laboratories for research purposes. Typical shipments usually involve 1,000–5,000 mosquito eggs with a total transit time of around 24–72 hours. They include imports of GM mosquito strains from partner laboratories (for instance, from Imperial College London) where the GM strains are generated by germline transformation.

In addition, there are imports of wild strains (from colonies established using field-caught mosquitoes) from partners in African countries (for instance, the Institut de Recherche en Sciences de la Santé or IRSS, in Burkina Faso). These are imported to test, in laboratory cages, the capabilities of the GM strains introgressed into the genetic background of wild mosquitoes from a potential future field site. This information is valuable in risk assessments and in planning for future field studies.

The pathway for technology development and testing taken by Polo GGB for Target Malaria is based on international recommendations for the use and testing of GM insects for population control [8]. It follows a step-by-step approach for efficacy and safety, first carrying out tests in contained laboratories in regions to which the target organisms are not endemic. As a consequence, it is necessary to move both wild and GM strains between different partner laboratories to perform the initial regulatory studies. The GM strains are then exported to contained-use facilities in destination countries for further regulatory studies.

Polo GGB exports both wild-type and GM strains to collaborators and partner laboratories in endemic countries, subject to specific permits from the relevant national authorities and intellectual property agreements set by the research partners. These movements may include exporting insects from wild-type strains that were originally collected in the field or GM mosquitoes based on those same background genotypes, after they have been approved for testing in confined settings or, eventually, for release into the field in the endemic country.

Documentation required for shipment

The shipment of live GM insects is subject to international standards as well as national regulations that are specific to each country. The documents required for shipping insects (e.g. mosquitoes) for research purposes specify whether the insects are considered an 'infectious substance' (i.e. are infected and can transmit a pathogen). If a GM insect is also classed as an infectious substance, this property takes priority over its having been genetically modified, and so these more rigorous requirements should be followed. Insects carrying pathogenic agents are subject to special rules and permits that will not be addressed in this article, as the author's experience is based on pathogen-free insects. Permits for the deliberate release of GM insects into the environment are also addressed by different regulations and are not discussed here.

The shipment of live insects requires liaison with the competent authorities (see below) responsible for authorising the importation and transit of such material. No transportation of the material can occur unless it is accompanied by a complete set of documents: the commercial document (pro-forma invoice), import permit and free export declaration.

A commercial document is required for all shipments and gives a complete and detailed description of the material shipped, including insect species of origin, category and quantity (in grams), the place of origin, reason for sending, harmonised tariff system code, and the complete name and address of the parties involved, including the country of the sender and the receiver (also known as the consignor and consignee/user) [9]. The value of the product should always be included in the commercial document. Customs rules in each country require valuation of the material and no country accepts a zero value. If the material has no commercial value, a symbolic amount is indicated (usually € 1 or US$ 1), with an indication that the material is for research purposes only.

Two prerequisites for shipping GM insects are the import permit and free export declaration. The import permit is requested by the receiver institute and obtained from the competent authority of the destination country, which also approves the import, transit, use and disposal of research samples under conditions that ensure that risks to public and animal health are controlled [9]. The competent authority designated to regulate GM organisms differs from country to country. In Italy, import permits are issued by the Ministry of Health and in the United States of America by the Animal and Plant Health Inspection Service of the US Department of Agriculture. In other countries, however, import permits are often issued by the Ministry of the Environment (or equivalent). Moving insects between institutions in the same country or between European Member States does not require an import permit.

The free export declaration states that the goods included in the shipment are not subject to export restrictions under current national regulations and international agreements. The shipper is legally certifying that no illicit or prohibited items are included in the package. In addition to the free export declaration, an export permit may be requested, to ensure that the material exported by a research institute complies with the provisions of international agreements. An export permit is issued by the competent authority of the country from which the material is shipped.

Since insects, especially mosquitoes, could serve as vectors for diseases and be infected with human or animal pathogens, a sanitary (health) certificate may be required at the discretion of the competent authority of the destination country to certify the non-contagious and non-infectious status of the shipped material. The sanitary certificate includes the information that the eggs (or other life stage for shipping) are free of known human or animal pathogens. There is some debate around the use of sanitary certificates for this purpose, as discussed in Wohlfarter *et al.* [10] and Quinlan *et al.* [11] in this issue. Ultimately, it is the responsibility of the sender institute to certify the non-infectious status of the insects. Genetically modified insects that have been reared and bred in laboratories for research purposes can be declared pathogen-free in the same way as wild-type insects. Thus there are no additional restrictions on the movement of GM insects or documentation required, compared to non-GM insects of the same species, unless requested by the countries involved in the trade.

Depending on the nature of the insect movement and the relationship between sending and receiving institutes, a material transfer agreement (MTA) or documentation of responsibilities and liabilities might be needed. Material transfer agreements are contracts that govern the relationship between the laboratories – including the rights and obligations of the recipient when it comes to the use and potential commercial exploitation of materials. Those agreements are of profound political importance in the context of forging greater collaborations in malaria control research (and other research), particularly when it comes to mediating economic and power imbalances between laboratories in the North and South. For example, an MTA could indicate that the receiving institute does not have the right to transfer the material further, or to register intellectual property from the material, or other such restrictions (see Sreerama Kumar *et al.* [12], this issue).

It is recommended, especially when shipping transgenic insects, to accompany the material with a certificate of analysis to ensure the recipient laboratories are informed of the nature of the genetic modifications. The certificate should include an exhaustive description of the modifications and their origin, instructions on how to rear and maintain the strain, and any protocols or other instructions to detect and verify the transgene (polymerase chain reaction assay, fluorescent markers, phenotype).

Other documents may be included in the shipment paperwork to give additional information on the ecological and health status of the material (a declaration of sensitive goods or declaration of non-dangerous goods) or to reduce the risk of material damage (exemption from X-ray checks).

A shipment may be stopped by customs officials due to incorrect or incomplete paperwork, leading to delays in shipments that can compromise the survival of the insects or other material and their usefulness in the destination country.

Shipping live mosquito eggs is always challenging and requires co-ordination between the sender, the logistics

providers, the courier and the recipient to ensure correct transport and timely arrival. Owing to the complexity of such movements, the author recommends using specialised logistics companies (e.g. World Courier, AmerisourceBergen) that can guide the parties through the preparation of the paperwork, including customs regulations, and are sensitive to the importance of transit time.

Packaging

Transport of GM insects (or GM organisms in general) is subject to strict national and international regulations. These regulations describe safety standards and packaging requirements, to ensure that the contents of the package do not escape or are not dispersed into the environment during handling or transportation. These regulations include:

- those of the International Air Transport Association (IATA) and International Civil Aviation Organization (ICAO) for transport by air [13];
- the European Agreement concerning the International Carriage of Dangerous Goods by Road and by Rail for transportation by road or rail in Europe and neighbouring countries [14, 15];
- the requirements of the International Maritime Organization for international shipping by sea [16].

The packaging must also prevent any handlers in the shipment chain, or any member of the public who could come into contact with the package, from being exposed to its contents. This is not only for safety, but to ensure the integrity of the material.

As a group, GM organisms are classified as dangerous goods when transported by air. Therefore, the applicable requirements are the IATA Dangerous Goods Regulations, classified as UN 3245 (Miscellaneous dangerous substances and articles, including environmentally hazardous substances) and Packing Instruction P904 (ICAO/IATA PI959 – shipping containment as Class 9) [13, 17].

Genetically modified organisms also need to be packaged to meet any biosecurity containment requirements during ordinary handling in transportation. This requires a triple packaging system (Figure 1), consisting of:

- the primary receptacle, which is a leak-proof sealed bag that contains the mosquito eggs;
- the secondary container, a leak-proof large plastic tube used to enclose and protect the primary receptacle;
- rigid outer packaging, into which the secondary container is placed, with suitable cushioning material. Outer packaging protects the contents from external effects, such as physical damage.

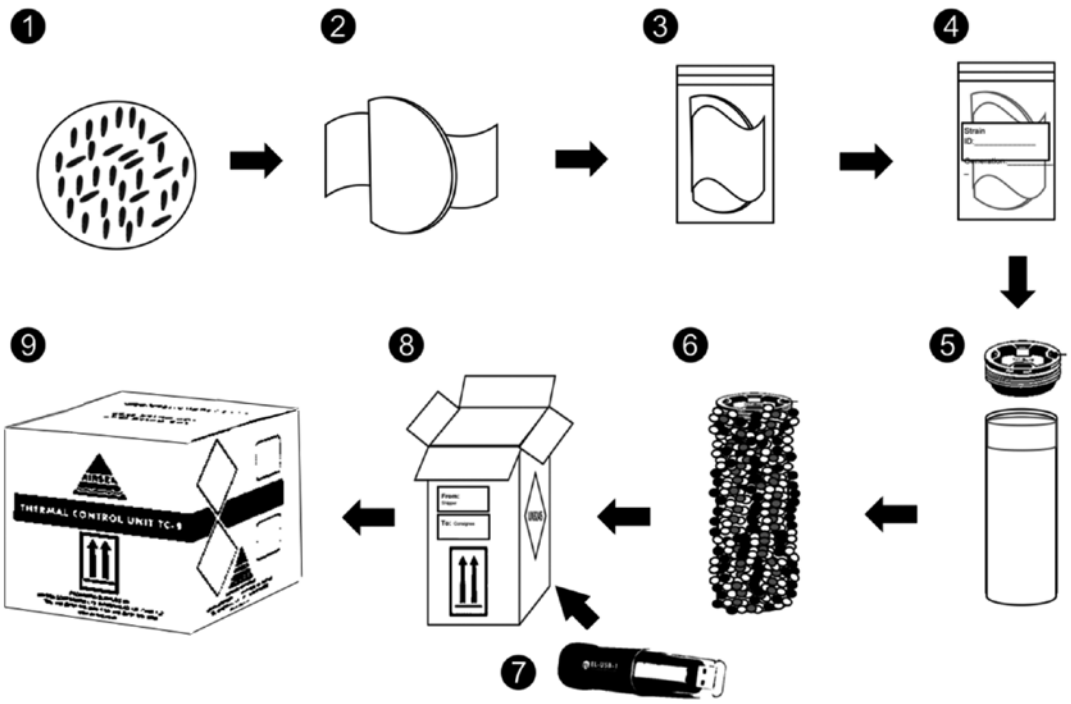

Figure 1

Shipment of live *Anopheles* mosquitoes

Non-motile forms, such as eggs or non-flying stages (1), are wrapped in wet filter paper (2) and sealed into a leak-proof primary container (3), properly labelled (4). A secondary insulated container (5) is used to protect the primary container. Outer packaging (6) protects the secondary container, which is inserted into rigid outer packaging (8), and labelled according to the type of biological material included. An environmental data logger could also be added (7) to monitor conditions during transport. Finally, the material and all required documentation are added to the parcel (9)

Additional precautions can be requested, for example, adding a parafilm or appropriate tape around the lid of the container to prevent any leakage.

The material to be transported must be labelled in such a manner as to notify any handler that the item to be transported contains GM organisms. For this reason, the label UN3245 must be displayed on opposite sides of the outer packaging on a background of a contrasting colour and must be clearly visible and legible.

Each layer of packaging must be of such construction and sturdiness as to independently prevent the release of the material under normal conditions during transport (temperature, humidity, pressure), and each layer must be independently closable or sealable.

It is recommended to ship mosquitoes as eggs or non-flying stages. This is because moving insects at a reproductively viable life stage may pose health risks, especially in countries where disease pathogens are present and known to be vectored by such insects. Handling and shipment of non-motile life stages reduces the risk of escape and also increases survival during transportation, since eggs are usually more resistant than adults.

However, the non-motile eggs need to be protected from extreme heat, cold and desiccation during transport, so insulated or adequate containers are necessary to maintain insect viability. If transportation is anticipated to be lengthy, the package may additionally include a cold pack to help regulate temperature. Most insects do not require large containers, feeding or added water during transportation (see the case described by Denton *et al.* [18], this issue, as an exception). However, depending on the species of mosquito (e.g. *Anopheles*), eggs may need to be kept wet for survival. Wrapping eggs in damp filter paper inside a sealed bag is usually sufficient, and no free water should be used to minimise the risk of leaking.

Movement of insects carrying gene drives

There are no regulations specific to the transport and movement of insects (or organisms in general) carrying gene-drive elements, and the documentation for gene-drive organisms is no different from that required for non-driving GM strains [19]. However, recommendations and guidelines for the movement of gene-drive organisms are in place to ensure that additional precautions are taken [5, 6, 7]. These include the recommendation to avoid shipping reproductively viable life stages, the use of rigorous packaging approved for shipment of hazardous organisms to ensure containment, and accompanying documentation of the responsibilities and liabilities of partners, as is the case for other GM organisms.

Contained-use approvals specify that the receiving laboratories recognise the invasive nature of the transferred insects and ensure that the destination laboratory is suited to receive and maintain GM organisms carrying gene drives. A certificate of analysis for the strain should be included with the shipment of gene-drive strains, and biosafety officers at the receiving institute should verify that containment conditions and authentication processes are adequate and that they are ready to receive such shipments [5, 6].

Conclusions

Exchanging live insects between institutions for research purposes is a very common practice, involving both wild-type and transgenic strains.

Strict regulations and packaging requirements are in place to ensure that live GM organisms are not accidentally dispersed into the environment and to protect handlers that could come into contact with the package. Despite these requirements, risk management for shipment of GM mosquitoes differs little from that associated with the same species unmodified. Transport of insects that can transmit human or animal diseases, such as mosquitoes, requires rigorous packaging, careful handling and appropriate documentation, especially to those countries with pathogens that could be vectored by the transported insects. The same is true for species that could become established in new environments, without authorisation for release.

When shipping mosquitoes carrying gene-drive elements, it is recommended to validate the strain using molecular procedures, both before and after transport, to ensure there are no unintended genotypes linked to the drive trait.

Regulations for the import and transit of GM insects for research differ from country to country, and recommendations currently exist to guide the researcher in ensuring proper transportation between laboratories. A key factor is that these shipments are destined for laboratory studies, not field releases, which have different requirements and ramifications.

Acknowledgements

Special thanks go to Tania Persampieri for her invaluable help and comments on the manuscript and for drafting the figure. The author also wishes to thank John Mumford, Megan Quinlan, John Connolly, Silke Fuchs and the anonymous reviewer(s) for useful comments on the manuscript. Target Malaria is supported by a grant from the Bill & Melinda Gates Foundation and Open Philanthropy.

Transferts d'insectes génétiquement modifiés à des fins de recherche

A. Simoni

Résumé

Les insectes jouent un rôle essentiel dans la recherche. Nombre de laboratoires mettent actuellement au point des technologies qui visent à lutter contre les insectes vecteurs de maladies ou ravageurs des cultures, en introduisant des modifications génétiques dont le but est de limiter la reproduction des insectes ou de les rendre résistants à la transmission des maladies. Ces outils font appel au guidage génétique, qui permet la transmission héréditaire de ces traits de manière durable et peu onéreuse. Dans le contexte actuel où les collaborations scientifiques au niveau international sont devenues systématiques, les transferts d'insectes génétiquement modifiés entre laboratoires de différents pays sont désormais très fréquents. L'auteur décrit les dispositions réglementaires et les directives applicables au transport d'insectes génétiquement modifiés à des fins de recherche. Il s'appuie sur l'expérience acquise par un laboratoire italien doté du statut de centre de recherche et qui, à ce titre, participe aux expéditions de moustiques sauvages et génétiquement modifiés dans le cadre d'un consortium scientifique international pour la mise au point d'outils durables de lutte contre le paludisme.

Mots-clés

Confinement d'arthropodes – Guidage génétique – Insectes – Insectes génétiquement modifiés – Transport.

Desplazamiento de insectos genéticamente modificados con fines de investigación

A. Simoni

Resumen

Los insectos son un elemento fundamental para la investigación. Numerosos laboratorios están desarrollando técnicas destinadas a combatir a insectos que actúan como vectores o constituyen plagas agrícolas empleando modificaciones genéticas que bien frenan la reproducción del insecto o bien confieren mayor resistencia a la transmisión de enfermedades. Entre tales herramientas figuran elementos de genética dirigida que posibilitan la propagación de esos rasgos genéticos de manera a la vez autosostenida y rentable. Teniendo en cuenta lo habituales que son hoy en día las iniciativas de investigación que entrañan colaboración internacional, el traslado de insectos genéticamente modificados entre laboratorios sujetos a diferentes jurisdicciones reglamentarias es también algo muy común. El autor expone los requisitos y directrices que rigen el transporte de insectos genéticamente modificados con fines de investigación, basándose en la experiencia de un laboratorio italiano que, como parte de un consorcio internacional de investigación que elabora herramientas sostenibles de lucha antipalúdica, cumple funciones de centro de investigación en cuanto al transporte de mosquitos salvajes o modificados.

Palabras clave

Contención de artrópodos – Genética dirigida – Insectos – Insectos genéticamente modificados – Transporte.

References

[1] Cook K.R. & Parks A.L. (2022). – The international exchange of *Drosophila melanogaster* strains. *In* Safety, regulatory and environmental issues related to international trade of live insects (J.D. Mumford & M.M. Quinlan, eds). *Rev Sci Tech*, **41** (1), 82–90. https://doi.org/10.20506/rst.41.1.3305

[2] Jinek M., Chylinski K., Fonfara I., Hauer M., Doudna J.A. & Charpentier E. (2012). – A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity. *Science*, **337** (6096), 816–821. https://doi.org/10.1126/science.1225829

[3] Alphey L.S., Crisanti A., Randazzo F. & Akbari O.S. (2020). – Standardizing the definition of gene drive. *Proc. Natl Acad. Sci. USA*, **117** (49), 30864–30867. https://doi.org/10.1073/pnas.2020417117

[4] Burt A. & Crisanti A. (2018). – Gene drive: evolved and synthetic. *ACS Chem. Biol.*, **13** (2), 343–346. https://doi.org/10.1021/acschembio.7b01031

[5] Benedict M.Q., Burt A., Capurro M.L., De Barro P., Handler A.M., Hayes K.R., Marshall J.M., Tabachnick W.J. & Adelman Z.N. (2018). – Recommendations for laboratory containment and management of gene drive systems in arthropods. *Vector Borne Zoonotic Dis.*, **18** (1), 2–13. https://doi.org/10.1089/vbz.2017.2121

[6] American Committee of Medical Entomology (ACME) & American Society of Tropical Medicine and Hygiene (ASTMH) (2022). – Containment practices for arthropods modified with engineered transgenes capable of gene drive. Addendum 1 to the Arthropod Containment Guidelines, Version 3.2., 2021. *Vector Borne Zoonotic Dis.*, **22** (1), 3–17. https://doi.org/10.1089/vbz.2021.0035

[7] Akbari O.S., Bellen H.J. [...] & Wildonger J. (2015). – Safeguarding gene drive experiments in the laboratory. *Science*, **349** (6251), 927–929. https://doi.org/10.1126/science.aac7932

[8] World Health Organization (WHO) (2021). – Guidance framework for testing genetically modified mosquitoes, 2nd Ed. WHO, Geneva, Switzerland, 196 pp. Available at: https://www.who.int/publications-detail-redirect/9789240025233 (accessed on 13 January 2022).

[9] European Union (EU) (2011). – Commission Regulation (EU) No. 142/2011 of 25 February 2011 implementing Regulation (EC) No. 1069/2009 of the European Parliament and of the Council laying down health rules as regards animal by-products and derived products not intended for human consumption and implementing Council Directive 97/78/EC as regards certain samples and items exempt from veterinary checks at the border under that directive text with EEA relevance. Document 32011R0142. *Off. J. Eur. Union*, **L 54**, 254 pp. Available at: https://eur-lex.europa.eu/eli/reg/2011/142/oj/eng (accessed on 13 January 2022).

[10] Wohlfarter M., Oliva C.F. & Nelson C. (2022). – Streamlining cross-border shipping of live invertebrates. *In* Safety, regulatory and environmental issues related to international trade of live insects (J.D. Mumford & M.M. Quinlan, eds). *Rev Sci Tech*, **41** (1), 178–190. https://doi.org/10.20506/rst.41.1.3315

[11] Quinlan M.M., Mumford J.D., Messori S., Enkerlin W.R., Shimura J., Smith L., Dass B., Oliva C.F., Nelson C., Chand R. & Torres G. (2022). – Issues and gaps in international guidance and national regulatory systems affecting international live insect trade. *In* Safety, regulatory and environmental issues related to international trade of live insects (J.D. Mumford & M.M. Quinlan, eds). *Rev Sci Tech*, **41** (1), 198–210. https://doi.org/10.20506/rst.41.1.3317

[12] Sreerama Kumar P., Sreedevi K., Amala U., Gupta A. & Verghese A. (2022). – Shipment of insects and related arthropods into and out of India for research or commercial purposes. *In* Safety, regulatory and environmental issues related to international trade of live insects (J.D. Mumford & M.M. Quinlan, eds). *Rev Sci Tech*, **41** (1), 158–164. https://doi.org/10.20506/rst.41.1.3312

[13] International Air Transport Association (IATA) (2021). – Infectious Substances Shipping Guidelines, 16th Ed. IATA, Montreal, Canada, 225 pp.

[14] United Nations Economic Commission for Europe (UNECE) (2021). – ADR 2021: Agreement concerning the International Carriage of Dangerous Goods by Road, Vol. 2. UNECE, Geneva, Switzerland, 638 pp. Available at: https://unece.org/sites/default/files/2021-01/ADR2021_Vol2e.pdf (accessed on 28 March 2022).

[15] Intergovernmental Organisation for International Carriage by Rail (OTIF) (2021). – Regulation concerning the International Carriage of Dangerous Goods by Rail. Appendix C. *In* Convention concerning International Carriage by Rail (COTIF). OTIF, Berne, Switzerland, 4 pp. Available at: https://otif.org/fileadmin/user_upload/otif_verlinkte_files/07_veroeff/03_erlaeut/06_Appendix_C.pdf (accessed on 4 March 2022).

[16] International Maritime Organization (IMO) (2022). – Annex 8: Resolution MSC.294(87) (adopted on 21 May 2010): Adoption of amendments to the International Maritime Dangerous Goods (IMDG) Code. IMO, London, United Kingdom, 86 pp. Available at: https://www.navcen.uscg.gov/pdf/marcomms/imo/msc_resolutions/MSC294(87).pdf (accessed on 28 March 2022).

[17] United Nations Economic Commission for Europe (UNECE) (2021). – Packing Instruction P904. *In* ADR 2021: Agreement concerning the International Carriage of Dangerous Goods by Road, Vol. 2. UNECE, Geneva, Switzerland, 126. Available at: https://unece.org/sites/default/files/2021-01/ADR2021_Vol2e.pdf (accessed on 28 March 2022).

[18] Denton J.A., Joubert D.A., Goundar A.A. & Gilles J.R.L. (2022). – International shipments of *Wolbachia*-infected mosquito eggs: towards the scaling-up of World Mosquito Program operations. *In* Safety, regulatory and environmental issues related to international trade of live insects (J.D. Mumford & M.M. Quinlan, eds). *Rev Sci Tech*, **41** (1), 91–99. https://doi.org/10.20506/rst.41.1.3306

[19] European Food Safety Authority (EFSA) Panel on Genetically Modified Organisms (GMO), Naegeli H. […] & Firbank L.G. (2020). – Adequacy and sufficiency evaluation of existing EFSA guidelines for the molecular characterisation, environmental risk assessment and post-market environmental monitoring of genetically modified insects containing engineered gene drives. *EFSA J.*, **18** (11), e06297. https://doi.org/10.2903/j.efsa.2020.6297

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